

The RMSD ceiling: an information limit on geometry-based transition-state metrics

Kfir Kaplan^{*1,3} and Alon Grinberg Dana^{†1,2,3}

¹Wolfson Department of Chemical Engineering, Technion – Israel Institute of Technology, Haifa 3200003, Israel

²Grand Technion Energy Program (GTEP), Technion – Israel Institute of Technology, Haifa 3200003, Israel

³Boeing-Technion SAF Innovation Center, Technion – Israel Institute of Technology, Haifa 3200003, Israel

Abstract

Machine-learning models for transition-state (TS) geometry prediction are almost universally benchmarked by the root-mean-square deviation (RMSD) of atomic positions, or by the mean absolute error of the interatomic-distance matrix (D-MAE), relative to a reference TS. We show that these Euclidean measures are unreliable proxies for the quantity a practitioner actually needs: whether a predicted geometry optimizes to a genuine TS at a target level of density functional theory. We optimized 520 interpolation-based guesses for 260 reactions from the Transition1x database [Schreiner M., *Sci. Data* 2022, 9, 779] at the r²SCAN-3c level. Every geometry-based quality measure – Kabsch-aligned RMSD, D-MAE, a reaction-coordinate projection, and cheap semi-empirical descriptors – saturates in a narrow band of the area under the receiver-operating-characteristic curve (AUC), 0.77–0.84, for predicting optimization success, and 18% of guesses within 0.28 Å of the reference fail nonetheless.

Many of those “failures” are not failures at all: 38% converge to a genuine but different first-order saddle point, and 77% of those lie lower in energy than the reference (median difference ~ 131 kJ mol⁻¹) – genuine alternative first-order saddles that a reference-matching metric penalizes as errors, though current automated structure perception cannot reliably tell which reaction each one belongs to. A controlled experiment identifies the cause. Displacing each reference saddle by a prescribed distance along different directions changes the probability of recovering it from 33% to 98%, so distance alone cannot determine the outcome; yet once direction is held fixed as well, the ceiling persists at AUC ≈ 0.83 . The saturation is therefore an information ceiling, not a shortcoming of any metric: the best geometric measure we construct, a parameter-free displacement-weighted RMSD, already operates at that limit.

The conclusions replicate on a second, chemically broader database, and on a state-of-the-art flow-matching generator, where oracle best-of- N selection roughly triples the deployable single-draw success rate yet no reference-free score we tried picks the good sample, so reported best-of- N figures overstate what a user obtains. Two consequences follow. A cheap GFN2-xTB force descriptor evaluated on the guess predicts convergence about as well as oracle RMSD and requires no reference structure – useful precisely because the ceiling leaves little room for a better one. And the field needs a change of yardstick: judging a predicted TS by whether it optimizes to any valid reactant-to-product saddle rather than by its RMSD to a single reference,

^{*}kfir.k4@gmail.com

[†]Corresponding author: alon@technion.ac.il

and, for stochastic generators, reporting the quality of a single draw or a reference-free-selected one rather than an oracle best-of- N .

1 Introduction

Predicting transition-state (TS) geometries is a central problem in computational reaction chemistry [1], and a rapidly growing set of machine-learning models now addresses it – from the early graph-based generator TS-GCN [2] through diffusion models that build the saddle from a 2D graph [3] or from 3D reactant and product structures [4], to the recent optimal-transport and flow-matching approaches [5, 6, 7]. The more recent of these are trained on the Transition1x database [8] and evaluated on held-out reactions from it. Almost without exception the headline benchmark for these models is geometric similarity to a reference TS – the root-mean-square deviation (RMSD) of atomic coordinates after optimal superposition, or the mean absolute error of the interatomic distance matrix (D-MAE).

Evaluation practice across this body of work is uniform in kind: the metric is almost always geometric, either a Cartesian RMSD to the reference saddle or the D-MAE. Which one a paper reports tracks its training data. Models trained on Transition1x headline a median RMSD to the held-out reference – 0.08 Å for OA-ReactDiff [4] and 0.053 Å for ReactOT [5] – and RitS [7] reports the lowest median of any generator evaluated on it. Models trained on the earlier Grambow and RDB7 reaction sets, such as TSDiff [3] and GoFlow [6], instead report the D-MAE or the coverage of their draws.

Many of these works already look past pure geometry. OA-ReactDiff pairs its generator with a confidence model that flags unreliable predictions, and both it and ReactOT report barrier errors of a few kcal mol⁻¹ with respect to their own DFT reference barriers – a measure of geometric fidelity to that reference, not of thermochemical accuracy. TS-GCN [2], TSDiff, and GoFlow each validate a fraction of their guesses physically, optimizing them to a saddle and checking that it connects the intended reactant and product. TSDiff and GoFlow even report that some of their apparent “errors” are saddles lower in energy than the reference.

Yet every headline score is still measured against a single reference saddle, and that reference is easy to over-trust. The Transition1x saddles were located by nudged-elastic-band searches at the ω B97x/6-31G(d) level [8], and none is guaranteed to be the lowest-energy saddle between its endpoints, or even the only one. Broader datasets reinforce the point: RDB7 [9] and RGD1 [10] span far more chemistry and tabulate full distributions of barrier heights, so any one collection samples only a slice of the accessible saddles. Its energetic coverage reflects how the dataset was built, not the reactions themselves.

The reason the field defaults to these Cartesian measures is practical, not principled. The physically correct test of a predicted TS – optimizing it to a first-order saddle and verifying, via an intrinsic reaction coordinate (IRC) [11], that it connects the intended reactant and product – requires Hessians and reaction-path integrations far too expensive to run for every prediction during model development. Machine-learned Hessians [12] and reactive interatomic potentials [13] lower this cost but do not remove the need for a target-level verification. A cheap geometric distance to a stored reference is an attractive surrogate. Implicit in its use is an assumption that, to our knowledge, has never been tested directly: that a higher similarity to the reference implies a more useful guess, i.e., one that is more likely to converge to a valid TS under a standard saddle-point optimizer.

An analogous reckoning has already taken place in structure-based drug design, where similarity to a single crystallographic pose was long the standard benchmark. The PoseBusters analysis [14]

Table 1: Failure-prediction AUC of RMSD under four per-atom weighting schemes ($n = 400$).

Weighting of the per-atom RMSD	AUC (failure)
uniform (plain Kabsch RMSD)	0.806
graph-distance decay from reactive atoms	0.819
Euclidean-distance decay from reactive atoms	0.820
displacement $ r^P - r^R $ (parameter-free)	0.839

showed that deep-learning docking methods tuned to minimize pose RMSD routinely generate physically invalid geometries, and the CASF (Comparative Assessment of Scoring Functions) benchmarks [15] established a distinction we return to for the generative model in Section 3.6: a model’s sampling power – placing a good candidate somewhere among its outputs – is separate from its *scoring power* – ranking that candidate first without access to the answer [16]. TS prediction, we will show, is subject to the same split.

Here we directly test that assumption – that a higher geometric similarity to the reference implies a more useful guess. We take deliberately simple, geometry-only predictors, optimize their guesses at a consistent density-functional-theory (DFT) level, and ask what – if anything – about a guess predicts whether it converges to a valid saddle, drawing conclusions only from quantities a practitioner can compute without already knowing the answer. Beneath the choice of error metric lies the following question: when the reference is only one of the many valid saddles a reaction can admit, does a guess’s resemblance to that single structure tell us whether it is useful at all? The field has thus already begun to look past pure geometry; what it lacks is a demonstration of whether geometric similarity is merely incomplete or fundamentally bounded, and of what a criterion should measure once it is. We provide both – and, for evaluations that must remain geometric, identify the strongest such metric: a parameter-free displacement-weighted RMSD that reaches the ceiling the standard RMSD and D-MAE fall short of.

2 Methods

2.1 Data and reference

We used the Transition1x database [8] of reactant, product and TS geometries (originally at the ω B97x/6-31G(d) level). To obtain an internally consistent reference at our working level, we re-optimized each Transition1x TS at the r^2 SCAN-3c composite DFT level [17] (ORCA 6.0.0 [18]). The set comprised 260 reactions (520 guesses); 259/260 reference saddles converged, all with exactly one imaginary vibrational frequency, confirming genuine first-order saddle points. Reactants and products were likewise optimized at r^2 SCAN-3c to define the minima used for connectivity. Areas under the curve (Tables 1 and 2, Fig. 1) are reported on the 200-reaction batch (400 guesses) for which the complete descriptor set, including the semi-empirical force and energy, was computed; the alternative-saddle census (Section 3.2) used all 260 reactions (520 guesses).

To test generalization beyond first-row organic chemistry, we repeated the whole protocol on a second database, Reaction-QM [19]. We used only its density-functional-optimized subset (B3LYP-D3/TZVP), from which we sampled 200 unimolecular reactions; these contain second- and third-row elements absent from Transition1x. All geometries were re-optimized at r^2 SCAN-3c exactly as above, so the two databases were compared at an identical working level and under an identical success criterion. Results are reported in Section 3.5.

2.2 Predictors

Two geometry-only predictors generated the guesses whose fate we studied: midpoint, the linear interpolation $\frac{1}{2}(D^R + D^P)$ of the reactant and product distance matrices, embedded in three dimensions by classical multidimensional scaling (MDS); and bond-order, which models the length of each reactive bond from Pauling’s bond-order/bond-length relation and interpolates the remaining pairs. These are not intended to be competitive with modern learned models; they are cheap, transparent sources of realistic guesses.

2.3 Downstream optimization and success

From each guess we ran a TS optimization with analytic Hessians (ORCA `OptTS`) followed by a vibrational analysis. A guess was counted a *success* if the optimization converged, the result had exactly one imaginary frequency, and its Kabsch-aligned RMSD to the reference saddle was below 0.3 Å. Optimization cost was recorded as the number of geometry cycles and the wall-clock time.

2.4 Metrics and their evaluation

We compared guess-quality measures both requiring the reference TS (Kabsch RMSD; D-MAE; a change-weighted D-MAE; a projection of the aligned error onto the reaction coordinate, split into components parallel and perpendicular to it; the un-normalized root-square deviation; and a reactive-atom-only RMSD) and deployable descriptors requiring only the guess and the reactant/product (the GFN2-xTB force norm on the guess, and its energy relative to the endpoints). GFN2-xTB [20] denotes the second-generation geometry-, frequency- and non-covalent- interaction-parametrized semi-empirical tight-binding method. Every measure used in this work is defined mathematically in Appendix A.

Because the outcome is binary (*success* or *failure*), we quantified how well a continuous measure separates the two classes by the area under the receiver-operating-characteristic curve (AUC). Intuitively, the AUC is the probability that a randomly chosen *failure* is assigned a larger error than a randomly chosen *success*: 0.5 is uninformative and 1.0 is perfect separation. The measure needs no decision threshold and is unchanged by any monotone rescaling of the metric.

2.5 Controlled displacement experiment

To test whether a metric’s ability to separate success from failure is limited by the metric itself or by the surface, we constructed guesses whose deviation was fixed by design rather than observed. Each reference saddle was re-optimized at the GFN2-xTB [20] level with the Sella saddle-point optimizer [21], and the Hessian was computed there by finite differences. Displacement directions were taken as the imaginary mode, the softest and stiffest internal vibrations, and an isotropic Gaussian vector – each Cartesian component drawn independently from a standard normal and the vector normalized, so that every orientation is equally likely. Because Kabsch alignment removes rigid-body motion, the six translation/rotation modes were projected out of every direction before use; nothing else was, so the isotropic vector was not orthogonalized against the imaginary mode and retains a component along it. Without this projection, a mode dominated by rotation produces a displacement an order of magnitude smaller than its nominal length. The step length was then solved by bisection so that the achieved Kabsch RMSD equaled the target; the achieved value was stored for every trial and checked during analysis. The saddle search was repeated from each displaced structure and counted as a return if it converged within 0.3 Å of the original xTB saddle. The experiment was attempted on all 200 reactions of each database. A reaction entered the

analysis only if its reference saddle relaxed to a converged GFN2-xTB saddle carrying exactly one negative curvature and yielding a usable set of vibrational modes; 187 of 200 Transition1x and 186 of 200 Reaction-QM reactions met this condition, the remainder failing the xTB saddle search, the Hessian, or the one-imaginary-mode test. Each surviving reaction contributed 16 trials – four displacements (0.2, 0.4, 0.6, 0.8 Å) along each of the four directions – for 2,992 and 2,976 trials respectively, at GFN2-xTB throughout so that no reference-level inconsistency entered. Because the isotropic direction was drawn at random, its seed was derived from the reaction identifier rather than from the process, so that splitting the batch across cluster jobs could not alter the result.

2.6 Generative model

To test whether a metric’s limited ability to separate success from failure is specific to interpolation-based guesses, we also generated saddles with RitS [7], a flow-matching model that predicts a TS from reactant and product structures. We used the released general checkpoint (not the Transition1x-specific one), which avoids train–test leakage on our Transition1x reactions; the reactant and product minima are the r²SCAN-3c geometries used everywhere else. Because RitS is stochastic, we drew ten samples per reaction and kept them all. The predicted atom ordering matches the input, which we verified per reaction. We then optimized two of the ten samples at r²SCAN-3c under the same success criterion as the other predictors: the first draw (what a single call returns) and the sample closest to the reference saddle (the oracle best-of-ten). Reference-free selection among the ten was evaluated with three scores – the geometric medoid, the lowest GFN2-xTB force norm, and the lowest GFN2-xTB energy – none of which sees the reference.

2.7 Connectivity

For each converged alternative saddle we computed the IRC – the minimum-energy descent path in both directions. IRC runs frequently terminated before reaching a minimum, so each endpoint was relaxed briefly at GFN2-xTB before perception. We identified the two endpoints by molecular-graph perception – covalent bond graphs and graph isomorphism as implemented in RMG [22, 23, 24], following the structure-perception approach of the Automated Rate Calculator (ARC) [25] – restricted where possible to the bonds that change across the reaction so that conformational differences were ignored. An endpoint was labeled reactant, product, or a distinct intermediate. Symmetry-equivalent artifacts (energy within 4.8 kJ mol^{−1} of the reference saddle) were removed before this analysis.

3 Results

3.1 Geometry metrics information ceiling

A geometric metric is only as useful as its ability to anticipate the outcome of an optimization, so we begin by defining that outcome precisely. Two distinct binary outcomes are available: *convergence*, whether the saddle search terminated at any first-order saddle at all, and *success*, the stricter requirement that it reached the reference saddle (within 0.3 Å, with exactly one imaginary frequency). All AUC values quoted below refer to *success* unless stated otherwise; discrimination of mere *convergence* is uniformly weaker (RMSD: AUC 0.733 against AUC 0.806 for *success*), because a guess may converge perfectly well to the “wrong” saddle (Section 3.2).

Table 2 reports how well each measure predicts optimization failure ($n = 400$). Kabsch RMSD is the best single measure at AUC 0.806; D-MAE is slightly lower (0.769). Every measure – including

the perpendicular reaction-coordinate error, which carries essentially all of RMSD’s signal while the parallel component (AUC 0.658) carries almost none – lands in a narrow band around AUC 0.80. The distributions overlap heavily: 30 of the 171 guesses within 0.28 Å of the reference fail (18%, over both batches) and successes reach large RMSD, so no distance threshold cleanly separates the classes (Fig. 2).

Two controls sharpen the point. The un-normalized root-square deviation ($\text{RMSD} \times \sqrt{N}$) gives AUC 0.808, statistically indistinguishable from RMSD: the $1/N$ size normalization carries no convergence information. A reactive-atom-only RMSD is *worse* (AUC 0.783): a hard binary restriction to the reacting center discards the whole-molecule fit that partly governs which basin the optimizer enters.

Between these extremes, a continuous weighting that concentrates on the reaction zone without discarding the rest improves on plain RMSD: in Table 1 the three weighted rows all exceed the “uniform” row, which is plain Kabsch RMSD at AUC 0.806. Weighting each atom’s squared deviation by its displacement across the reaction, $w_i = |\mathbf{r}_i^P - \mathbf{r}_i^R|$ (Kabsch-aligned), needs no free parameter and no explicit reaction center and reaches AUC 0.839; exponential decays in graph-bond distance and in Euclidean distance from the reactive atoms give AUC 0.819 and 0.820. Thus the best purely geometric convergence measure is neither global RMSD nor a reactive-atom mask, but this motion-weighted RMSD: a parameter-free reweighting that lifts discrimination above plain RMSD, and well above D-MAE (AUC 0.769), to essentially the ceiling. Where a reference-based geometric score must be reported, it is therefore a better benchmark metric than the RMSD and D-MAE now standard in the literature – though, as the rest of this section shows, still a bounded one. Finally, a random forest [26] trained on purely deployable features (GFN2-xTB force and energy plus reactant/product geometry, with no reference TS) reaches AUC 0.791 – matching oracle RMSD without ever seeing the answer.

The limit is also not an artifact of the coordinate system. Repeating the analysis in redundant internal coordinates – the representation the optimizer itself uses – does not help: a reaction-change-weighted deviation over all bonds, angles and dihedrals (each expressed as an equivalent atomic displacement) reaches only AUC 0.590, because the many soft, periodic dihedral coordinates flood the metric; restricted to bonds and angles it recovers to AUC 0.740, still below Cartesian RMSD. Weighting the coordinate types by the optimizer’s own default initial-Hessian stiffnesses (0.5, 0.2, 0.1 for stretches, bends and torsions, respectively) as a harmonic-energy metric does not help either (AUC 0.610): the per-coordinate downweighting is far too small to offset how much further the dihedrals swing. Their typical deviation is some five times a bond’s, so after squaring they still carry over 90% of the summed harmonic contribution. The result is unchanged when the metric is built on ORCA’s exact redundant internal coordinates, parsed from the optimization logs rather than reconstructed (AUC 0.600 against 0.590 for our bond-graph set). Added to the displacement-weighted RMSD, internal-coordinate deviations raise the cross-validated AUC from 0.839 to 0.844, within the fold-to-fold noise (± 0.02). Internal coordinates – even the optimizer’s own DOFs, weighting, and representation – thus add nothing to the Cartesian metric’s power to anticipate whether a guess converges to a saddle.

The limit is not specific to geometric information either. An electronic descriptor – the deviation of the guess’s Mayer bond orders [27, 28] from the reference, read from the same DFT output – reaches AUC 0.700. Added to the displacement-weighted RMSD, it again raises the cross-validated AUC only within noise (0.839 to 0.844). Features computed from the reactant and product alone – their geometric separation, reaction energy, and total bond reorganization – predict convergence more weakly still. The best of them, the $R \rightarrow P$ displacement, reaches only AUC 0.700, and the reaction energy is uninformative (AUC 0.460). These describe how demanding a reaction is on average, not whether a particular guess lies in a saddle basin, and so cannot exceed the guess-based

ceiling.

The limit is not an artifact of choosing a single metric. A random forest over all eight geometric measures (the five geometric rows of Table 2 and the three weighted variants of Table 1) reaches AUC 0.829, no higher than the single displacement-weighted RMSD, and adding the deployable xTB force does not improve on it either. Every guess-based measure we tried – geometric or semi-empirical, single or ensembled, oracle or deployable – lands in a narrow band that ceilings near AUC 0.84 (Fig. 1). None of the guess-based descriptors we examined, whether single, weighted, or ensembled, separates the classes further; the residual variation tracks the potential-energy surface rather than the guess.

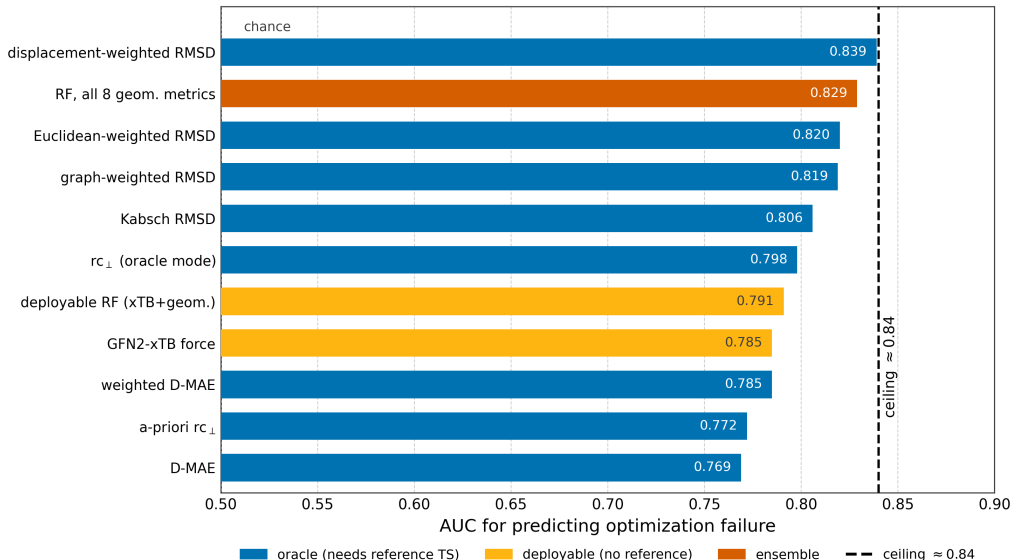


Figure 1: None of the guess-based measures predict optimization success beyond $AUC \approx 0.84$ ($n = 400$). Oracle measures (blue) require the reference TS; deployable descriptors (gold) require only the guess and the reactant/product; the random-forest ensemble (red) combines all eight geometric measures. None exceeds the single displacement-weighted RMSD.

We attribute the surprising transferability – of a semi-empirical force to the r^2 SCAN-3c optimization outcome – to the breadth of the TS basin: what governs convergence is not sub-Ångström geometric accuracy but whether the guess lies within the basin of attraction of the saddle, a region tens of degrees and tenths of an Ångström wide. GFN2-xTB [20] reproduces this coarse curvature and gradient structure well even where it mis-estimates the barrier itself, so a small semi-empirical residual force is a faithful indicator that the higher-level optimizer will fall into the same basin. The force norm thus measures proximity to a stationary point on a correlated surface, which is exactly the quantity that predicts convergence.

3.2 Alternative saddles among the “failures”

Of the 239 failures among the 520 guesses of both batches, 142 (59%) did not converge, 92 (38%) converged to a first-order saddle other than the reference, and 5 collapsed to a minimum. For the alternative saddles, the median magnitude of the energy difference from the reference, $|E_{\text{alt}} - E_{\text{ref}}|$, is 131 kJ mol^{-1} , and 77% are lower in energy; only 13% lie within 4.8 kJ mol^{-1} of the reference (symmetry-equivalent artifacts). These are bare r^2 SCAN-3c electronic energies, and we do not

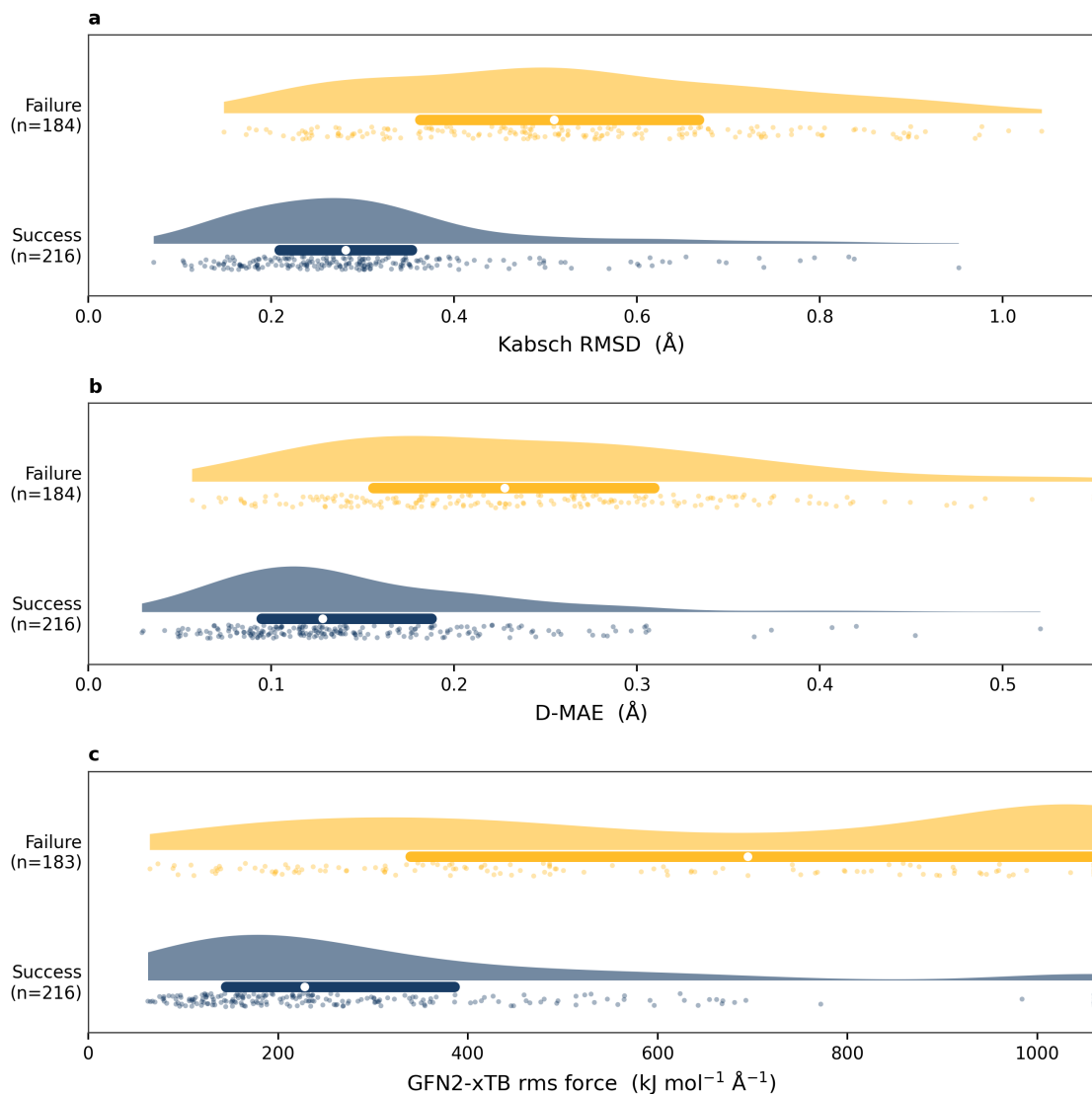


Figure 2: Rainclouds (half-violin, box with white median, and per-guess points) comparing the successful ($n = 216$) and failed ($n = 184$) classes of the 400 batch-2 guesses, for three representative guess-quality measures: (a) Kabsch RMSD, (b) D-MAE, and (c) the deployable GFN2-xTB rms force on the guess, for which one failed guess lacks an xTB single point ($n = 183$). The split is on *success* as defined in Section 3.1, not on mere convergence. Values beyond the axis range are drawn at the right-hand limit. Class medians shift in the expected direction while the distributions overlap across most of their range.

Table 2: Failure-prediction AUC of geometric and semi-empirical guess-quality measures, with and without access to the reference TS.

Measure	AUC (failure)	Needs the reference TS?
Kabsch RMSD	0.806	yes
D-MAE	0.769	yes
Change-weighted D-MAE	0.785	yes
Reaction coord. $\perp$ error	0.798	yes
Reaction coord. $\parallel$ error	0.658	yes
GFN2-xTB force (guess)	0.785	no
Deployable random forest	0.791	no

offer them as accurate barrier heights: a composite DFT method is chosen here for consistent geometries and Hessians at tractable cost, not for thermochemistry, and no zero-point, thermal, or higher-level correction is applied. The claim they support is the ordinal one – that the alternative saddle is a different stationary point, usually the lower one – and the gaps involved are an order of magnitude larger than the errors such a method is expected to make. A third of what the metric calls failure is therefore the optimizer locating a genuine, usually more stable, first-order saddle that the reference database simply does not contain. This is consistent with reports from the generative-model literature itself: TSDiff [3], for instance, was observed to sample TSs with lower barrier heights than the reference database, which its RMSD-based evaluation scores as deviations rather than discoveries. What these saddles are chemically – and why that question is harder to answer than it looks – is taken up in Section 3.7.

The magnitude matters kinetically: a median discrepancy of ~ 131 kJ mol $^{-1}$ in barrier height is far larger than the zero-point and entropic terms that a bare electronic energy omits, so at 298 K the two saddles are of decisively different kinetic relevance [29]. The comparison rests on the further assumption that the two saddles belong to the same reaction. Where an alternative saddle is instead a step of a different mechanism, the comparison is between rates of different reactions and the prefactors need not cancel. Even then, a barrier gap of this size dwarfs any plausible prefactor variation of a few orders of magnitude: a benchmark that treats these saddles as errors is not penalizing a small geometric slip but discarding TSs of an entirely different kinetic weight – precisely the regime where getting the right saddle, rather than one close in RMSD, is decisive.

3.3 Robustness of the ceiling

The class overlap of Fig. 2 could be an artifact of the setup rather than a real limit: the predictor might be too weak, the binary outcome too coarse, or the metric confounded with the reaction itself. Three controls rule out each in turn. A better guess generator does not lift the ceiling. The bond-order predictor marginally outperforms the midpoint baseline (success 56% vs 52%; 20.0 vs 21.3 optimization cycles, paired), but the effect is small, consistent with guess quality not being the limiting factor.

Convergence is binary, but its cost – the number of geometry-optimization cycles – is graded, and offers a second, more forgiving test of the metrics. Over the guesses that do converge, the geometry measures correlate with the cycle count only weakly to moderately (Spearman ρ : RMSD 0.59, perpendicular reaction-coordinate error 0.55, weighted D-MAE 0.54, D-MAE 0.51; the cheap GFN2-xTB force reaches 0.55). The same ordering as for success recurs, at a lower ceiling: a better guess is somewhat cheaper to optimize, but geometry explains well under half the rank variance

in cost. A perfect guess (the reference TS itself) converges in ~ 5 cycles against ~ 20 for the interpolated guesses, so the potential cost saving is large – yet not predictable from Cartesian similarity.

Restricting to low-RMSD guesses ($< 0.28 \text{ \AA}$) – those that geometric similarity alone would deem good enough, 171 of the 520 guesses of both batches – and re-testing the reactant- and product-derived features of Section 3.1, no feature separates the 30 failures from the 141 successes (all AUC 0.5–0.6). Reaction-amplitude features that appear predictive over all guesses (AUC 0.65–0.68) collapse to noise once RMSD is held fixed – they predict failure only by making the guess worse, not independently. The residual failures are structurally indistinguishable from the successes – the same class overlap seen across all guesses in Fig. 2; their cause lies in the potential-energy surface (competing saddles, basins), not in the reaction’s gross structure.

3.4 Why the ceiling exists: a controlled displacement experiment

The above analysis indicates a correlational relationship: guesses differ from the reference saddle in the size and the direction of their displacement at once, so a metric’s failure could reflect a poor metric or a poor predictor. We separate the two by constructing displacements whose distance is fixed by design. From each reference saddle – first relaxed to a GFN2-xTB saddle so that the experiment is internally consistent – we displace the geometry by a prescribed Kabsch RMSD along four directions and repeat the saddle search: the imaginary reaction mode, the softest real vibration, the stiffest vibration, and an isotropic Gaussian direction. The six rigid-body translation and rotation components are projected out of every direction, and the step length is solved numerically, so every trial sits at the requested distance to better than $10^{-3} \text{ \AA}$; the achieved distance is recorded for every trial and audited. A trial “returns” if the search converges back to the original xTB saddle point within $0.3 \text{ \AA}$. Because these displacements are drawn from a small set of prescribed lengths, almost every pair of trials is tied in distance; AUC values are therefore computed with mid-ranks, without which ties are broken arbitrarily and the statistic becomes meaningless.

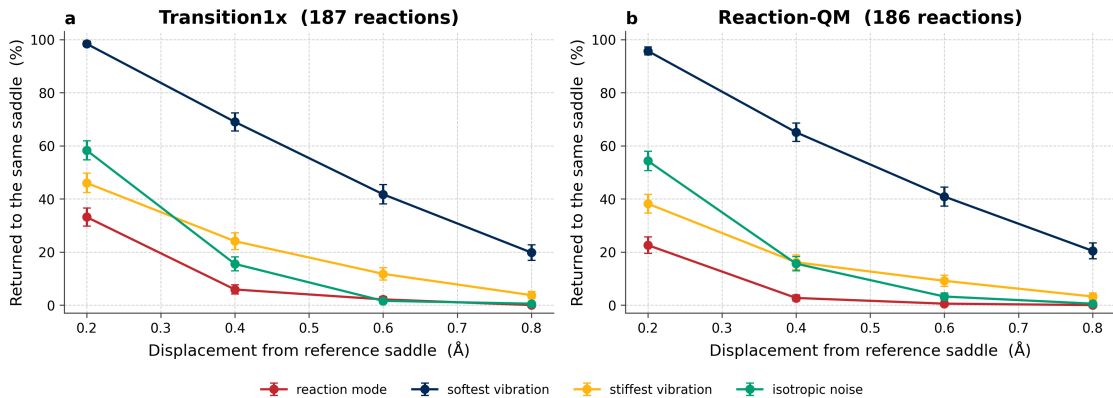


Figure 3: Return rates to the original saddle from controlled displacements, for (a) Transition1x (187 reactions, 2,992 trials) and (b) Reaction-QM (186 reactions, 2,976 trials) – the reactions of each 200-reaction batch whose reference saddle relaxed to a genuine GFN2-xTB saddle (Section 2.5), each contributing 16 trials. Points with the same displacement from the reference saddle differ only in the direction of that displacement, so the vertical spread is due to direction alone. Error bars are binomial standard errors.

At a fixed displacement of $0.20 \text{ \AA}$, the return rate ranges from 33% along the reaction mode to

98% along the softest vibration on Transition1x, and from 23% to 96% on Reaction-QM (Fig. 3); direction-independence is rejected at every displacement in both databases (χ^2 on the 4×2 table of direction against return, 3 d.o.f., $p < 10^{-14}$ throughout). The ordering is physically sensible and identical across the two databases: the reaction mode is the unstable direction and leaves the basin fastest, while the softest vibration is the most forgiving. This anisotropy is what the topology of a saddle region predicts – reaction paths curve, branch, and pass through valley-ridge inflections [30], so displacement along the unstable eigenvector escapes the basin far sooner than an equal step along a soft orthogonal mode. At a fixed distance, direction alone swings the return rate from 33% to 98%. Cartesian distance therefore does not determine the outcome, and no measure built on distance alone can.

Holding direction fixed as well recovers about 0.06 in AUC. Pooled over directions, the AUC for displacement predicting failure to return is 0.775 (Transition1x) and 0.757 (Reaction-QM); restricted to a single direction it rises to 0.831 and 0.823. Direction-blindness is thus a real but modest penalty, and far from the whole gap to a perfect predictor. The residue is the point: even with the distance controlled to 10^{-3} Å, the direction held fixed, and the reference saddle identical, displacement predicts return only at $\text{AUC} \approx 0.83$. That is essentially the value reached by the best purely geometric measure on real predictions (displacement-weighted RMSD, AUC 0.839, Table 1).

The anisotropy could still be an artifact of the semi-empirical surface. We tested that directly, repeating the experiment at the working level of theory: 320 additional r²SCAN-3c saddle optimizations on 40 Transition1x reactions, with the displacement directions taken from the reference DFT Hessians rather than recomputed. The direction effect reproduces. At 0.20 Å the return rate is 98% along the softest vibration against 28% along the reaction mode ($p = 3 \times 10^{-8}$ and 8×10^{-14} at the two displacements).

The two levels agree quantitatively as well. Compared like for like – the same two displacements, and the 36 of these 40 reactions that also cleared the xTB screen of Section 2.5 – their AUCs differ by at most 0.02: pooled over directions, 0.651 at r²SCAN-3c against 0.667 at GFN2-xTB; within a fixed direction, 0.711 against 0.723. The semi-empirical surface is therefore a faithful proxy here, and the larger xTB sample can carry the statistical weight. These absolute values are lower than those quoted above, but only because the comparison rests on two displacements instead of four. That is the range effect of Section 3.5 again: restricting the same xTB data to those two displacements moves the within-direction AUC from 0.831 to 0.709.

Two independent routes therefore arrive at the same number. This is the mechanism. The ceiling is not an artifact of metric design, nor merely of RMSD’s blindness to direction, nor of noise in the predictors: it survives an experiment in which all three are eliminated or controlled by construction. What remains is variation between reactions in how large a basin their saddle possesses, a property of the potential energy surface that none of the guess-based measures examined here accesses. Metrics saturate near AUC 0.84 because that is where the information these descriptors extract from a guess geometry is exhausted, and the best of them – the displacement-weighted RMSD, at AUC 0.839 – already operates at that limit.

Observations in this work cluster by reaction: each reaction contributes two guesses to the 400-guess analysis and 16 trials to the displacement experiment. The binomial standard errors and the contingency test reported above treat trials as independent and are therefore optimistic. The direction effect is far too large to be produced by that dependence, but the smaller contrasts – in particular the 0.033 gap in AUC between displacement-weighted and plain RMSD – should be read as provisional until reaction-level resampling is applied.

A corollary bears on how synthetic guesses should be built. Isotropic Gaussian noise, the natural way to manufacture guesses of a prescribed quality, does not behave like a typical direction: it tracks the reaction mode arm, the harshest of the four. Perturbing a reference structure isotropically

therefore understates the success rate a real predictor would achieve at the same nominal deviation, and is not a neutral surrogate for predictor error.

3.5 Generalization

All results above derive from a single database of small first-row organic reactions. To test whether they are an artifact of that chemistry, we repeated the entire protocol on 200 unimolecular reactions drawn from the density-functional subset of Reaction-QM [19], which extends to second- and third-row elements (Si, P, S, Cl) and larger systems. Reference saddles behave as before (197/200 converged, all with exactly one imaginary frequency), so the comparison is not confounded by a poor reference.

Guess quality falls, downstream success collapses, and more of the converged guesses land on an alternative saddle (Fig. 4). First, the predictors degrade sharply out of distribution: the median guess deviation rises from 0.35 Å to 1.02 Å, and only 2 of 400 guesses fall below 0.3 Å (against 155 of 400 on Transition1x). Second, downstream success collapses from 54% to 6%, with convergence falling from 73% to 34% and the median cost of a converged optimization rising from ~ 20 to ~ 32 cycles. Third, the fraction of converged guesses that reach an alternative saddle rather than the reference one grows, from 19% to 26%: as reactions become more complex, the reference TS accounts for a smaller share of the accessible saddles, reinforcing Section 3.7. Reaction-QM is harder in two ways at once – larger systems and heavier elements – and this design does not separate them; what follows turns on whether reference similarity transfers as a metric, not on which of the two makes the chemistry harder.

What does not change is the central negative result. Taken over the full range, the AUC for the deviation measure appears to *improve* to 0.920, which would suggest the ceiling of Section 3.1 had been broken. It has not. That figure is an artifact of range: discrimination is trivially easier when the guesses span a wide quality range (0.74–1.33 Å interquartile) than a narrow one (0.25–0.54 Å). Restricting the Reaction-QM guesses to the Transition1x interquartile range removes the advantage entirely – the AUC falls to 0.610 (95% bootstrap confidence interval 0.42–0.80, $n = 53$), statistically indistinguishable from both the Transition1x AUC of 0.730 (0.66–0.81) and from chance. Geometric similarity to the reference TS is therefore no more predictive on a harder and chemically broader database than on the one the conclusion was drawn from; if anything the mapping from deviation to outcome is itself database-dependent, so a threshold calibrated on one dataset does not transfer to another. The one result that does replicate is the predictor ordering: the bond-order predictor again edges out the midpoint baseline (7% vs 5% success), the same direction and the same negligible magnitude as before.

3.6 The ceiling in a generative model

The predictors studied so far are deliberately simple, so the $\text{AUC} \approx 0.84$ ceiling on predicting the optimization outcome from the guess might be a property of those weak, interpolation-based guesses rather than of the problem, and would lift for a modern machine-learned generator. We therefore repeated the analysis with RitS [7], a recent flow-matching TS generator that takes the reactant and product geometries and emits a predicted saddle. We use the authors’ released checkpoint, which is not trained on Transition1x, so our evaluation carries no train–test leakage. Two limitations of the tool reduce our coverage. It supports only the elements C, H, N, O, S, F, P, which removes 96 of the 200 Reaction-QM reactions, and its bond perception from Cartesian input fails outright on a further 16 Transition1x and 7 Reaction-QM reactions. The analysis therefore covers 184 of 200 Transition1x and 97 of 200 Reaction-QM reactions.

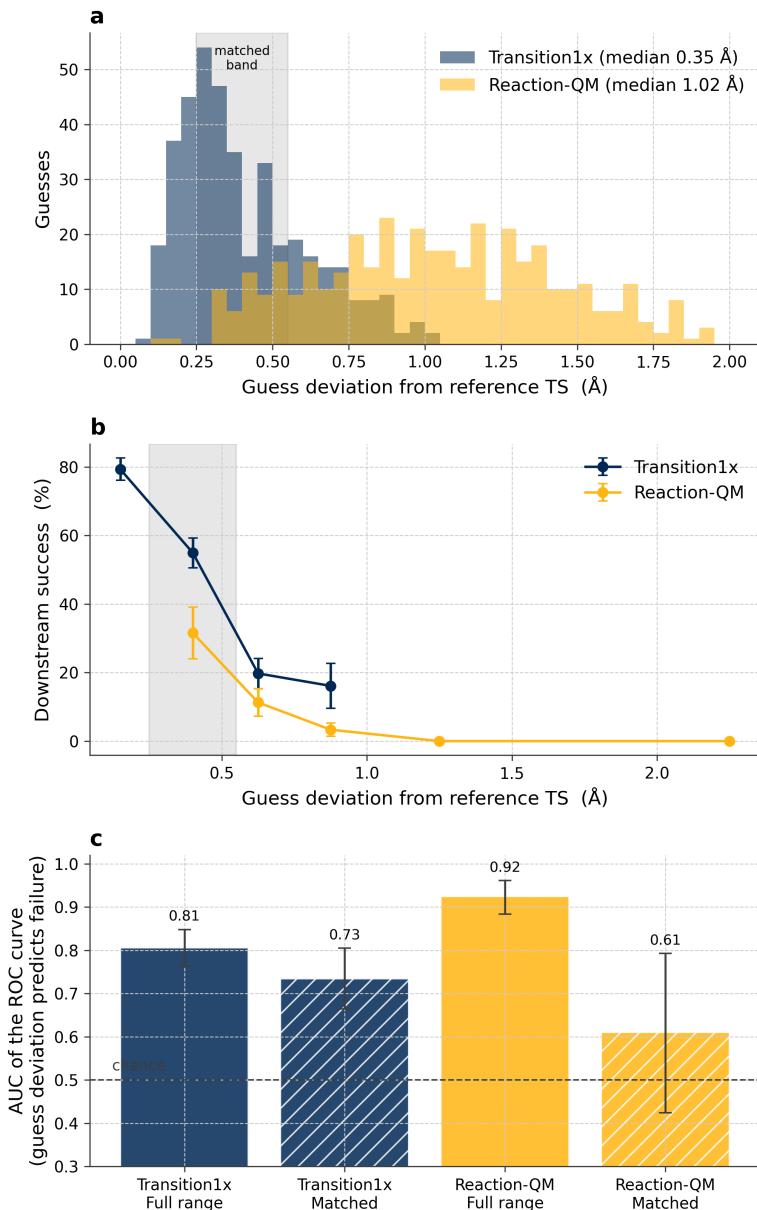


Figure 4: Generalization to Reaction-QM. Both databases are represented by the same 400 guesses per database as everywhere else in this work – two interpolation-based guesses (linear midpoint and bond-order interpolation, Section 2.2) for each of 200 reactions, optimized at r^2 SCAN-3c under the same success criterion. (a) The predictors degrade sharply out of distribution; the shaded band is the Transition1x interquartile range of guess deviation, used to match the two datasets. (b) Downstream success against guess deviation: at every matched deviation Reaction-QM succeeds less often, so a threshold calibrated on one database does not transfer to the other. (c) Area under the receiver-operating-characteristic curve (AUC) for guess deviation predicting failure – the probability that a randomly chosen failed guess is assigned a larger deviation than a randomly chosen successful one, so that 0.5 is chance (Section 2.4). The apparent improvement in discrimination on Reaction-QM (AUC 0.920) is an artifact of its much wider deviation range; restricted to the matched band it falls to AUC 0.610, statistically indistinguishable from both the Transition1x value and from chance. Error bars are binomial standard errors in (b) and 95% bootstrap confidence intervals in (c).

RitS is stochastic: each call draws an independent sample, and the ten draws we took per reaction span a median range of 0.67 Å on Transition1x (Fig. 5a) – wider than the entire spread between the methods compared in the rest of this paper. This makes the choice of what to report decisive. The single draw a user actually obtains has a median deviation of 0.82 Å (Transition1x) and 1.08 Å (Reaction-QM), no better than the linear-interpolation baseline. The best of ten draws, selected by proximity to the reference saddle, has a median deviation of 0.37 and 0.73 Å – a two- to threefold improvement, and the number a best-of- N benchmark would report.

That best-of- N number is an oracle. Selecting the closest sample requires the reference TS, which is exactly the unknown the model exists to predict. The gap it opens is not academic: optimized at r²SCAN-3c under our success criterion, the deployable single draw reaches the reference saddle for 16% (Transition1x) and 4% (Reaction-QM) of reactions, while the oracle best-of-ten reaches it for 44% and 23% (Fig. 5c; McNemar $p < 10^{-3}$ on both, discordant pairs 54 vs 2 and 18 vs 0). Oracle selection roughly triples the success rate – a gain entirely unavailable at prediction time.

The obvious question is whether a reference-free score can recover it. We tried three: the consensus (medoid) sample, which requires no electronic structure; the lowest GFN2-xTB force, which is the deployable descriptor this paper recommends elsewhere; and the lowest GFN2-xTB energy. None of the three selects a sample closer to the reference than chance: across both databases the sample each picks sits at a mean rank of 4.5–5.7 among the ten (Fig. 5b), against a chance expectation of 5.5 and the oracle’s 1.0; the consensus sample is marginally better than chance and the energy and force selectors are not. We did not optimize the selected samples, so this establishes that the selectors cannot reproduce the oracle’s proximity to the reference, not that they cannot reproduce its downstream success – a distinction this paper’s own argument demands, and one that requires DFT optimizations. What can be said is that the near-threefold best-of- N advantage lives in knowing which sample is closest to an answer one does not have, and that no cheap proxy we tried stands in for that knowledge. The information ceiling of Section 3.1 thus reappears for a modern generator in a sharper form – as a selection problem – and it cautions that best-of- N scores in the generative-TS literature overstate deployable performance by roughly a factor of three.

3.7 The nature of the alternative saddles

We attempted to classify each alternative saddle by computing its IRC at r²SCAN-3c and identifying the two endpoints (after a short GFN2-xTB relaxation of the frequently truncated IRC endpoints). Reference-TS controls, whose IRCs must connect their reactant and product, validated the endpoint extraction (6/6). Symmetry-equivalent artifacts (energy within 4.8 kJ mol⁻¹ of the reference) account for ~13% of the alternative saddles and were removed.

However, assigning the remaining endpoints to a chemical species proved not robust. Four reasonable identity criteria – covalent adjacency graph, InChI [31], graph isomorphism as implemented in RMG (as used by the Automated Rate Calculator, ARC [25]), and canonical SMILES [32] – returned between 6% and 91% “same reaction” on the identical set of endpoints. The spread is not noise between near-equivalent tests. Holding the perception step fixed – the same reactive-bond graph on the same relaxed endpoints – and varying only what is demanded of it reproduces most of it: requiring the two endpoints to reproduce the reference reactant and product *as a pair* scores 6%, while asking only whether each endpoint resembles *either* of them scores 88%. The criterion, not the chemistry, sets the answer.

Manual inspection identified three specific algorithmic blind spots behind that sensitivity, all aggravated by the open-shell character of many Transition1x species. First, *radical-center and multiplicity assignment* is not consistent across two geometries of the same molecule: perception

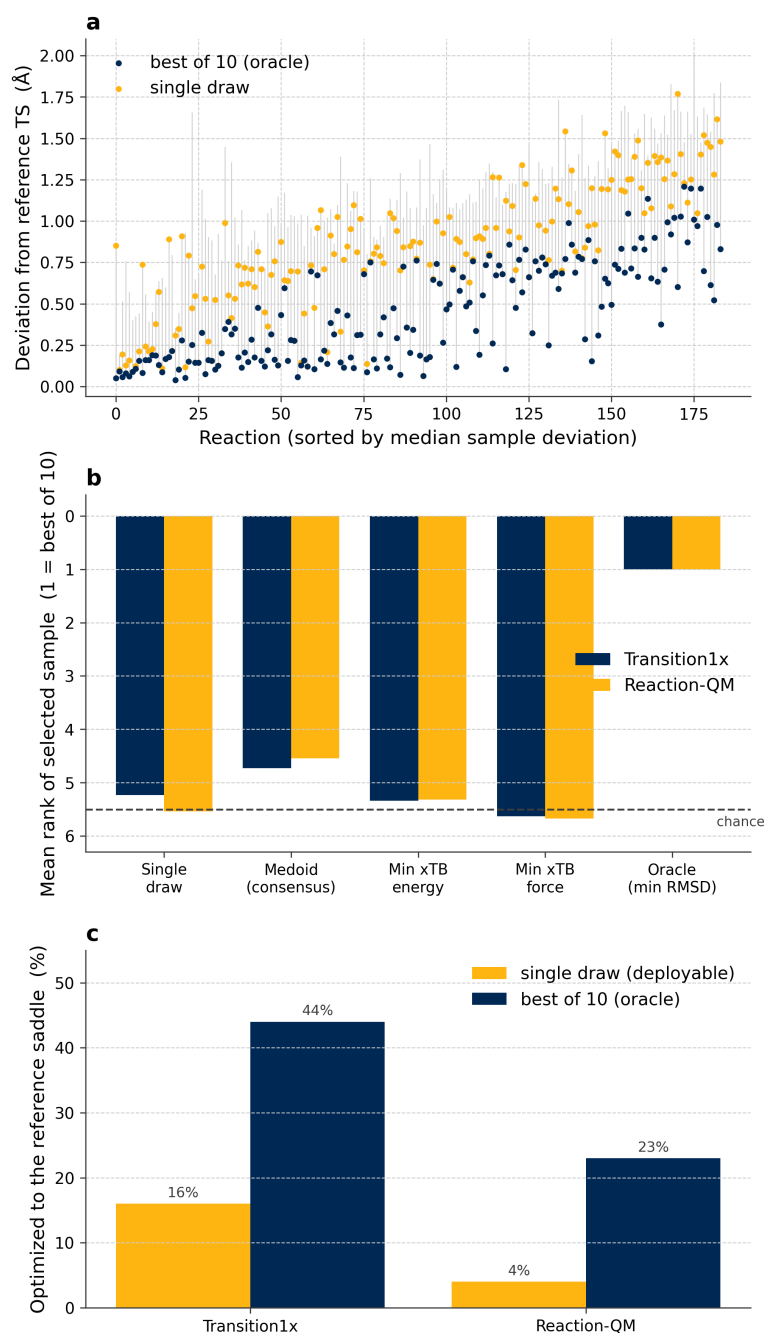


Figure 5: A state-of-the-art generator under the same lens. (a) RitS is stochastic: for each Transition1x reaction (sorted by median), ten draws span a wide range; the oracle best (blue) sits far below the single draw (orange). (b) Mean rank of the sample chosen by each reference-free selector, out of ten (1 = best); every deployable selector lands near the chance line, only the oracle (min RMSD to the reference) reaches 1. (c) Optimized at $r^2\text{SCAN-3c}$, the oracle best-of-ten roughly triples the deployable single-draw success on both databases; the reference-free selections were not optimized and do not appear here.

routines placed unpaired electrons on different atoms and returned different multiplicities for a species and its own relaxed copy. Second, *resonance-structure selection* is arbitrary – a delocalized radical is represented by one of several equivalent Kekulé/resonance forms, and graph isomorphism over Lewis structures then declares two forms of the same species distinct, while canonical SMILES and InChI do not, unless resonance forms are generated and filtered first [33]. Third, *path directionality* is lost: an IRC endpoint may relax to a conformer or tautomer of the reactant/product that the reference minimum does not match, and the two IRC branches occasionally invert, so reactant and product labels swap. The net effect is that a single species is repeatedly split into spurious “intermediates” – the same class of artifact that lets automated workflows admit species which cannot exist [34].

We therefore cannot state with confidence whether the alternative saddles are concerted alternatives of the same reaction, genuine stepwise elementary steps, or saddles of different reactions. This is itself a finding: for open-shell reaction networks, even the seemingly simple question “did the guess find the intended reaction?” is not answerable by current automated structure perception, and a robust, perception-independent connectivity criterion is an open problem. What we can state solidly, the foregoing notwithstanding, rests on energetics alone: the alternative saddles exist, are usually lower in energy, and are penalized by a reference-matching metric.

4 Discussion

RMSD and D-MAE measure proximity to *one* recorded saddle, whereas what a practitioner needs is that the guess lie in the basin of attraction of *a* valid saddle connecting reactant and product – a topological property. The two notions coincide only when the reference is the unique relevant saddle, which we find fails roughly a third of the time. Interpolation guesses routinely optimize to lower-energy saddles (77% of the alternative-saddle cases) absent from Transition1x, so a geometrically “wrong” prediction frequently reaches a more stable TS. Benchmarking against a single reference therefore rewards a model for reproducing the database’s particular saddle choice rather than for locating a valid, possibly lower-energy, saddle – even though, for these open-shell endpoints, we cannot robustly determine whether each lower saddle belongs to the same reaction, a stepwise variant, or a different one (Section 3.7).

The mechanism lies in the optimizer. A quasi-Newton saddle-point optimizer only ever sees the local Hessian; from a given guess it walks to whichever saddle the local curvature indicates, which need not be the closest in RMSD nor the reference. This is why none of the guess geometry functions we tested predicts success beyond AUC $\approx$ 0.84: the outcome is governed by the surface more than by the guess. The controlled-displacement experiment makes this concrete – direction at fixed distance swings the outcome from 33% to 98%, and even the direction-controlled ceiling matches the best real metric – so the saturation is an information limit, not a modeling artifact.

For generative models the ceiling reappears as a selection problem. A stochastic generator can place a good saddle among its samples far more often than any single draw suggests: for RitS the oracle best of ten roughly triples the deployable success rate. But choosing that sample requires the reference, and none of the reference-free scores we tried – consensus, xTB force, xTB energy – selects better than chance. This is the sampling-power/scoring-power split that structure-based drug design formalized a decade ago [16, 15]: generating a good candidate somewhere in the top- N is a different, easier problem than ranking it first without an oracle, and the gap between them is a general bottleneck of generative models, not a quirk of RitS. Best-of- N evaluation therefore measures the wrong quantity: it credits a model for a sample it cannot identify without the answer, and overstates deployable performance by about threefold. The figure to report is the expected

quality of a single draw, or of a draw selected by a genuinely reference-free rule. The distinction grows more consequential as these generators are handed to autonomous agents that construct kinetic models without a human in the loop [35], since such a pipeline has no reference against which to recognize the good draw.

Four changes to evaluation practice follow. Report convergence rate and IRC connectivity rather than RMSD/D-MAE; score against any valid reactant→product saddle rather than a single reference; for stochastic generators, report single-draw or reference-free-selected quality rather than oracle best-of- N ; and explore topological or energy-derivative-free training objectives – a force-norm or saddle-character penalty, for instance – that reward residence in a saddle basin over proximity to a particular geometry. For a cheap convergence proxy at scale, a GFN2-xTB force descriptor on the guess is about as informative as oracle RMSD and needs no reference structure.

5 Conclusion

Geometric similarity to a reference TS does not predict whether a predicted geometry is useful: every geometry metric saturates in a narrow band below AUC 0.84, and a large share of apparent failures are lower-energy saddles the reference database omits and the metric penalizes. A controlled experiment locates the cause. At a fixed distance from the saddle the outcome varies from 33% to 98% depending only on the direction of displacement, so distance cannot determine convergence; yet once direction is also controlled the ceiling persists at AUC $\approx$ 0.83, and the effect reproduces at r²SCAN-3c. The saturation is therefore not a deficiency of any particular metric but a limit on what these measures extract from a guess geometry: reactions differ widely in the size of the basin around their saddle, and none of the descriptors examined here captures that difference. This holds on a second, chemically broader database: once the comparison is matched for guess quality, geometric similarity is no more predictive there than on the database the conclusion was drawn from, and the mapping from deviation to outcome does not transfer between the two. The same limit constrains a state-of-the-art generative model: its oracle best-of- N score, roughly triple its deployable single-draw success, rests on a proximity to the reference that no reference-free rule we tried can identify, so reported best-of- N figures do not reflect what a user obtains. TS prediction should be benchmarked by surface behavior – convergence, and where determinable, connectivity – not by Cartesian distance, and generative models by the quality of a single or reference-free-selected draw rather than an oracle best-of- N . That determining connectivity is itself hard for open-shell reaction networks only strengthens the case that geometric distance is the wrong yardstick.

Acknowledgments

This research was supported by the Boeing–Technion SAF Innovation Center funded by the Boeing Company, by the Israel Science Foundation under Grant No. 185/23, and by the Stephen and Nancy Grand Technion Energy Program (GTEP). K.K. is grateful for the financial support from the Israeli Sustainable Aviation Fuel Knowledge Center (iSAF) and the Ministry of Innovation, Science and Technology.

Data and Software Availability Statement

All data and software needed to reproduce this work are openly available. The archive contains the reaction sets drawn from Transition1x and Reaction-QM; the guess geometries produced by each predictor; the reduced results of every r²SCAN-3c optimization, frequency and IRC calculation,

of the GFN2-xTB controlled-displacement experiment, and of the RitS samples, each as compact JSON keyed by reaction and predictor; and the analysis and plotting code that turns those files into every number, table and figure in this article, including the job-generation scripts used to run the DFT calculations on the cluster. Software used and not redistributed here is available from its authors: ORCA 6.0.0, GFN2-xTB through `tblite`, Sella, RMG-Py, RDKit and RitS. The archive is deposited at Zenodo under DOI 10.5281/zenodo.21954584; the manuscript source is at <https://github.com/DanaResearchGroup/ts-metrics-paper>.

Appendix

A Definitions of the metrics

Throughout, a structure of N atoms is a set of Cartesian coordinates $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N \subset \mathbb{R}^3$. We write $\mathbf{X}^g$ for the guess, $\mathbf{X}^\ddagger$ for the reference saddle, and $\mathbf{X}^R, \mathbf{X}^P$ for the reactant and product minima, all at r2SCAN-3c and in a fixed atom ordering.

A.1 Alignment and the per-atom deviation

Let $\mathcal{K}(\mathbf{X}; \mathbf{Y})$ denote the Kabsch superposition of $\mathbf{X}$ onto $\mathbf{Y}$: the rigid motion $\mathbf{x} \mapsto \mathbf{R}(\mathbf{x} - \bar{\mathbf{x}}) + \bar{\mathbf{y}}$ whose rotation $\mathbf{R} \in SO(3)$ minimizes $\sum_i \|\cdot - \mathbf{y}_i\|^2$. The per-atom deviation of a guess from the reference is

$$\mathbf{d}_i = [\mathcal{K}(\mathbf{X}^g; \mathbf{X}^\ddagger)]_i - \mathbf{x}_i^\ddagger, \quad \boldsymbol{\delta} = (\mathbf{d}_1, \dots, \mathbf{d}_N) \in \mathbb{R}^{3N}. \quad (1)$$

Because the multidimensional-scaling embedding fixes chirality arbitrarily, the mirror image $\mathbf{X}^g \cdot \text{diag}(-1, 1, 1)$ is also aligned and whichever of the two gives the smaller RMSD is used; for an achiral reaction the two describe the same saddle. Every Cartesian measure below is built from the same $\{\mathbf{d}_i\}$, so differences between them are differences of weighting, not of alignment.

A.2 Cartesian measures

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_i \|\mathbf{d}_i\|^2}, \quad (2)$$

$$\text{RSD (un-normalized)} = \sqrt{\sum_i \|\mathbf{d}_i\|^2} = \sqrt{N} \text{RMSD}, \quad (3)$$

$$\text{RMSD}_{\mathcal{A}} = \sqrt{\frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \|\mathbf{d}_i\|^2} \quad (\text{reactive atoms only}). \quad (4)$$

The reactive set $\mathcal{A}$ contains every atom whose row of the covalent adjacency matrix differs between reactant and product, $\mathcal{A} = \{i : \exists j, b_{ij}^R \neq b_{ij}^P\}$, with $\mathcal{A} = \{1, \dots, N\}$ in the degenerate case $\mathcal{A} = \emptyset$.

The weighted family (Table 1) is

$$\text{RMSD}_w = \sqrt{\frac{\sum_i w_i \|\mathbf{d}_i\|^2}{\sum_i w_i}}, \quad (5)$$

with the four weightings

$$\text{uniform: } w_i = 1, \quad (6)$$

$$\text{graph decay: } w_i = e^{-g_i/\tau}, \quad g_i = \min_{a \in \mathcal{A}} d_{\text{graph}}(i, a), \quad (7)$$

$$\text{Euclidean decay: } w_i = e^{-e_i/\tau}, \quad e_i = \min_{a \in \mathcal{A}} \|\mathbf{x}_i^R - \mathbf{x}_a^R\|, \quad (8)$$

$$\text{displacement: } w_i = \|\mathcal{K}(\mathbf{X}^P; \mathbf{X}^R)_i - \mathbf{x}_i^R\|. \quad (9)$$

Here d_{graph} is the bond-count distance in the reactant covalent graph and $\tau = 2$ (bonds, and Å respectively). Only the displacement weighting (9) is free of both a decay constant and an explicit reactive-atom set, which is the sense in which it is called parameter-free; it is the measure that reaches AUC 0.839.

A.3 Distance-matrix measures

With $D_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$ (rotation- and translation-invariant, so no alignment is needed),

$$\text{D-MAE} = \frac{2}{N(N-1)} \sum_{i < j} |D_{ij}^g - D_{ij}^\ddagger|, \quad (10)$$

$$\text{D-MAE}_w = \frac{\sum_{i < j} w_{ij} |D_{ij}^g - D_{ij}^\ddagger|}{\sum_{i < j} w_{ij}}, \quad w_{ij} = |D_{ij}^P - D_{ij}^R|, \quad (11)$$

the change-weighted form giving near-zero weight to spectator pairs without an arbitrary active/inactive threshold.

A.4 Reaction-coordinate projection

Let $\hat{\mathbf{v}} \in \mathbb{R}^{3N}$ be the normalized imaginary-frequency eigenvector of the reference saddle. The aligned error splits as

$$e_{\parallel} = |\boldsymbol{\delta} \cdot \hat{\mathbf{v}}|, \quad e_{\perp} = \|\boldsymbol{\delta} - (\boldsymbol{\delta} \cdot \hat{\mathbf{v}}) \hat{\mathbf{v}}\|, \quad e_{\parallel}^2 + e_{\perp}^2 = N \text{RMSD}^2. \quad (12)$$

Both are norms of the full $3N$ -vector in Ångström, i.e. not divided by $\sqrt{N}$.

A.5 Internal-coordinate deviation

Redundant internals q – bonds r_{ij} , angles θ_{ijk} , dihedrals ϕ_{ijkl} – are enumerated from the union of the reactant and product bond graphs, so forming and breaking bonds are included. Angular deviations are converted to an equivalent atomic displacement by the local bond length (Wilson- B principle), so that all coordinates share units of Ångström:

$$\Delta_q = \begin{cases} |r_{ij}^g - r_{ij}^\ddagger|, & \text{bonds,} \\ r_{ij}^\ddagger |\theta^g - \theta^\ddagger|, & \text{angles,} \\ r_{kl}^\ddagger |\text{wrap}(\phi^g - \phi^\ddagger)|, & \text{dihedrals,} \end{cases} \quad w_q = |q^P - q^R|, \quad (13)$$

with $\text{wrap}(x) = ((x + \pi) \bmod 2\pi) - \pi$ and dihedrals discarded when either defining angle lies within 20° of 0 or π . The parameter-free internal metric pools all coordinate types into a single change-weighted RMS, $\sqrt{\sum_q w_q \Delta_q^2 / \sum_q w_q}$, in exact analogy with (5). The harmonic-energy variant instead forms per-type RMS values M_{bond} , M_{ang} , M_{dih} and combines them as $\sum_t c_t M_t$ with the optimizer’s default initial-Hessian stiffnesses $(c_{\text{bond}}, c_{\text{ang}}, c_{\text{dih}}) = (0.5, 0.2, 0.1)$.

A.6 Electronic deviation: Mayer bond orders

With B_{ij} the Mayer bond order [27, 28] of the pair (i, j) – taken from the first Mayer block of the guess optimization log and the last block of the converged reference, reactant and product – the

oracle electronic metric is the change-weighted RMS deviation

$$M_{\text{MBO}} = \sqrt{\frac{\sum_{i < j} w_{ij} (B_{ij}^g - B_{ij}^\dagger)^2}{\sum_{i < j} w_{ij}}}, \quad w_{ij} = |B_{ij}^P - B_{ij}^R|, \quad (14)$$

pairs absent from a printed block (order below 0.1) being treated as $B_{ij} = 0$.

A.7 Deployable descriptors

These use only the guess and the two endpoints. With GFN2-xTB forces $\mathbf{F}_i$ and energy E evaluated at the guess geometry,

$$F_{\text{max}} = \max_{i,\alpha} |F_{i\alpha}|, \quad F_{\text{rms}} = \sqrt{\frac{1}{3N} \sum_{i,\alpha} F_{i\alpha}^2}, \quad \Delta E_{\uparrow} = E^g - \max(E^R, E^P), \quad (15)$$

together with $\Delta E_{gR} = E^g - E^R$ and $\Delta E_{gP} = E^g - E^P$. The force norm reported in Table 2 is F_{max} . The deployable random forest additionally uses reference-free geometric features: $\text{RMSD}(g, R)$, $\text{RMSD}(g, P)$, their absolute difference, $\text{RMSD}(R, P)$, the corresponding D-MAEs, N , and the fraction of interatomic distances of the guess lying outside the interval spanned by the endpoints, $|\{(i, j) : D_{ij}^g \notin [\min(D_{ij}^R, D_{ij}^P) - \epsilon, \max(D_{ij}^R, D_{ij}^P) + \epsilon]\}| / \binom{N}{2}$ with $\epsilon = 0.1$ Å.

A.8 Success and the area under the ROC curve

A guess is a *success* if the saddle optimization converges, the resulting structure has exactly one imaginary frequency, and its Kabsch RMSD to the reference saddle is below 0.3 Å; *convergence* drops the third condition. For a measure m evaluated on the success set $\mathcal{S}$ and the failure set $\mathcal{F}$, the reported area under the ROC curve is the Mann–Whitney statistic with mid-ranks,

$$\text{AUC} = \frac{1}{|\mathcal{S}||\mathcal{F}|} \sum_{s \in \mathcal{S}} \sum_{f \in \mathcal{F}} \left[\mathbf{1}(m_f > m_s) + \frac{1}{2} \mathbf{1}(m_f = m_s) \right], \quad (16)$$

i.e. the probability that a randomly drawn failure is assigned a larger error than a randomly drawn success, with ties counted as one half. The tie term is what makes the statistic meaningful in the controlled-displacement experiment (Section 3.4), where the prescribed step lengths tie almost every pair of trials. AUC is invariant under any strictly monotone rescaling of m and needs no decision threshold, which is why measures with incommensurable units – Ångström, bond orders, forces – can be compared on one axis.

References

- [1] Alon Grinberg Dana, Kevin M. Van Geem, Carlo Cavallotti, and William H. Green. Predictive chemical kinetic modeling: where we succeed, where we struggle, and what comes next. *ACS Engineering Au*, 6(1):1–19, 2026.
- [2] Lagnajit Pattanaik, John B. Ingraham, Colin A. Grambow, and William H. Green. Generating transition states of isomerization reactions with deep learning. *Physical Chemistry Chemical Physics*, 22(41):23618–23626, 2020.
- [3] Seonghwan Kim, Jeheon Woo, and Woo Youn Kim. Diffusion-based generative AI for exploring transition states from 2D molecular graphs. *Nature Communications*, 15:341, 2024.

- [4] Chenru Duan, Yuanqi Du, Haojun Jia, and Heather J. Kulik. Accurate transition state generation with an object-aware equivariant elementary reaction diffusion model. *Nature Computational Science*, 3(12):1045–1055, 2023.
- [5] Chenru Duan, Guan-Horng Liu, Yuanqi Du, Tianrong Chen, Qiyuan Zhao, Haojun Jia, Carla P. Gomes, Evangelos A. Theodorou, and Heather J. Kulik. Optimal transport for generating transition states in chemical reactions. *Nature Machine Intelligence*, 7:615–626, 2025.
- [6] Leonard Galustian, Konstantin Mark, Johannes Karwounopoulos, Maximilian P.-P. Kovar, and Esther Heid. GoFlow: efficient transition state geometry prediction with flow matching and E(3)-equivariant neural networks. *Digital Discovery*, 4(12):3492–3501, 2025.
- [7] Filipp Nikitin, Dylan M. Anstine, and Olexandr Isayev. Right into the saddle: Stereochemistry-aware generation of molecular transition states. *ChemRxiv*, 2026.
- [8] Mathias Schreiner, Arghya Bhowmik, Tejs Vegge, Jonas Busk, and Ole Winther. Transition1x – a dataset for building generalizable reactive machine learning potentials. *Scientific Data*, 9:779, 2022.
- [9] Colin A. Grambow, Lagnajit Pattanaik, and William H. Green. Reactants, products, and transition states of elementary chemical reactions based on quantum chemistry. *Scientific Data*, 7:137, 2020.
- [10] Qiyuan Zhao, Sai Mahit Vaddadi, Michael Woulfe, Lawal A. Ogunfowora, Sanjay S. Garimella, Olexandr Isayev, and Brett M. Savoie. Comprehensive exploration of graphically defined reaction spaces. *Scientific Data*, 10:145, 2023.
- [11] Kenichi Fukui. The path of chemical reactions – the IRC approach. *Accounts of Chemical Research*, 14(12):363–368, 1981.
- [12] Eric C.-Y. Yuan, Anup Kumar, Xingyi Guan, Eric D. Hermes, Andrew S. Rosen, Judit Zádor, Teresa Head-Gordon, and Samuel M. Blau. Analytical ab initio hessian from a deep learning potential for transition state optimization. *Nature Communications*, 15:8865, 2024.
- [13] Jonas Busk, Mikkel N. Schmidt, Ole Winther, Tejs Vegge, and Peter Bjørn Jørgensen. Graph neural network interatomic potential ensembles with calibrated aleatoric and epistemic uncertainty on energy and forces. *Physical Chemistry Chemical Physics*, 25:25828–25837, 2023.
- [14] Martin Buttenschoen, Garrett M. Morris, and Charlotte M. Deane. PoseBusters: AI-based docking methods fail to generate physically valid poses or generalise to novel sequences. *Chemical Science*, 15(9):3130–3139, 2024.
- [15] Minyi Su, Qifan Yang, Yu Du, Guoqin Feng, Zhihai Liu, Yan Li, and Renxiao Wang. Comparative assessment of scoring functions: The CASF-2016 update. *Journal of Chemical Information and Modeling*, 59(2):895–913, 2019.
- [16] Zhe Wang, Huiyong Sun, Xiaojun Yao, Dan Li, Lei Xu, Youyong Li, Sheng Tian, and Tingjun Hou. Comprehensive evaluation of ten docking programs on a diverse set of protein–ligand complexes: the prediction accuracy of sampling power and scoring power. *Physical Chemistry Chemical Physics*, 18(18):12964–12975, 2016.

- [17] Stefan Grimme, Andreas Hansen, Sebastian Ehlert, and Jan-Michael Mewes. r²scan-3c: A “swiss army knife” composite electronic-structure method. *The Journal of Chemical Physics*, 154(6):064103, 2021.
- [18] Frank Neese. Software update: The ORCA program system – version 6.0. *WIREs Computational Molecular Science*, 2025.
- [19] Kyunghoon Lee, Sungwoo Kang, Beomgyu Kang, Won-Joon Son, Seungmin Lee, Dae Sin Kim, and Woo Youn Kim. A comprehensive dataset of chemical reactions covering second and third row elements with million-scale quantum chemical calculations. *Scientific Data*, 13(983), 2026.
- [20] Christoph Bannwarth, Sebastian Ehlert, and Stefan Grimme. GFN2-xTB – an accurate and broadly parametrized self-consistent tight-binding quantum chemical method with multipole electrostatics and density-dependent dispersion contributions. *Journal of Chemical Theory and Computation*, 15(3):1652–1671, 2019.
- [21] Eric D. Hermes, Khachik Sargsyan, Habib N. Najm, and Judit Zádor. Sella, an open-source automation-friendly molecular saddle point optimizer. *Journal of Chemical Theory and Computation*, 18(11):6974–6988, 2022.
- [22] Mengjie Liu, Alon Grinberg Dana, Matthew S. Johnson, Mark J. Goldman, Agnes Jocher, A. Mark Payne, Colin A. Grambow, Kehang Han, Nathan W. Yee, Emily J. Mazeau, Katrin Blondal, Richard H. West, C. Franklin Goldsmith, and William H. Green. Reaction mechanism generator v3.0: Advances in automatic mechanism generation. *Journal of Chemical Information and Modeling*, 61(6):2686–2696, 2021.
- [23] Jackson W. Burns, Kirk Badger, Katrin Blöndal, Yunsie Chung, Anna Doner, Xiaorui Dong, Sevy Harris, Nora Khalil, Bonhyeok Koo, Shih-Cheng Li, Danielle Lucey, Nathan Morgan, Hao-Wei Pang, Timo T. Pekkanen, Seth G. Porter, Kevin Spiekermann, Su Sun, Julian Ufert, Yen-Ting Wang, Haoyang Wu, Chao Xu, Akshat Shirish Zalte, Jonathan W. Zheng, Alon Grinberg Dana, C. Franklin Goldsmith, William H. Green, Matthew S. Johnson, Bjarne Kreitz, and Richard H. West. RMG-Py v4: 17 years of open source reaction mechanism generation. *ChemRxiv*, 2026.
- [24] Matthew S. Johnson, Xiaorui Dong, Alon Grinberg Dana, Yunsie Chung, David Farina, Ryan J. Gillis, Mengjie Liu, Nathan W. Yee, Katrin Blondal, Emily Mazeau, Colin A. Grambow, A. Mark Payne, Kevin A. Spiekermann, Hao-Wei Pang, C. Franklin Goldsmith, Richard H. West, and William H. Green. RMG database for chemical property prediction. *Journal of Chemical Information and Modeling*, 62(20):4906–4915, 2022.
- [25] Alon Grinberg Dana et al. ARC – automated rate calculator. Zenodo, 2019.
- [26] Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001.
- [27] I. Mayer. Charge, bond order and valence in the ab initio SCF theory. *Chemical Physics Letters*, 97(3):270–274, 1983.
- [28] I. Mayer. Bond order and valence indices: A personal account. *Journal of Computational Chemistry*, 28(1):204–221, 2007.
- [29] Henry Eyring. The activated complex in chemical reactions. *The Journal of Chemical Physics*, 3(2):107–115, 1935.

- [30] Satoshi Maeda, Yu Harabuchi, Yuriko Ono, Tetsuya Taketsugu, and Keiji Morokuma. Intrinsic reaction coordinate: Calculation, bifurcation, and automated search. *International Journal of Quantum Chemistry*, 115(5):258–269, 2015.
- [31] Stephen R. Heller, Alan McNaught, Igor Pletnev, Stephen Stein, and Dmitrii Tchekhovskoi. InChI, the IUPAC international chemical identifier. *Journal of Cheminformatics*, 7:23, 2015.
- [32] David Weininger. SMILES, a chemical language and information system. 1. introduction to methodology and encoding rules. *Journal of Chemical Information and Computer Sciences*, 28(1):31–36, 1988.
- [33] Alon Grinberg Dana, Mengjie Liu, and William H. Green. Automated chemical resonance generation and structure filtration for kinetic modeling. *International Journal of Chemical Kinetics*, 51(10):760–776, 2019.
- [34] Nelly Mitnik, Sharon Haba, and Alon Grinberg Dana. Non-physical species in chemical kinetic models: A case study of diazenyl hydroxy and diazenyl peroxide. *ChemPhysChem*, 23(23):e202200373, 2022.
- [35] Alon Grinberg Dana. The agentic age of predictive chemical kinetics. *Chemical Science*, 17(1):27–35, 2026.